

Disposition for an incoming patient message

A three-bucket prototype with versioned physician review

Brandon Dent, MD

Former EM assistant community professor at UNR

<http://localhost:4121/stripped>

Local demo. Requires the local server.

48/50 with revised physician labels; original result retained at 44/49.

The score changed after physician reassessment. Model outputs and prompt are unchanged.

C22: incomplete ankle assessment



Exact patient message

24M. I rolled my ankle playing basketball last night. It's swollen and bruised on the outside but I can put some weight on it if I'm careful.

Fable: SELF_CARE

Physician: ASYNC_PHYSICIAN

Discussion images. Not part of the scored C22 input.

False negative: required clinician review missed

Potential harm: delayed evaluation of an injury that could include a fracture.

C22's text does not establish Ottawa criteria or a fracture diagnosis. Ottawa criteria are discussed in appendix 16.

C47: persistent sleep difficulty needs review

Fable SELF_CARE → physician ASYNC_PHYSICIAN

45F. I've had trouble falling asleep for the last month or so. Work has been stressful. I'm functioning but tired all day.

Persistent symptoms and daytime fatigue warrant clinician review under the physician reference.

The message does not assess mood or suicidality concerns.

False negative: required clinician review missed

Potential harm: delayed assessment of persistent insomnia and underlying causes.

Both Astra settings selected ASYNC_PHYSICIAN. No patient outcome or observed harm was measured.

Prior architecture: results and roadmap



V25: 27/50 completed releases; 21/49 exact-route agreement.

Next: test the simple baseline on unseen cases. Add one component only if it reduces clinically important misses.

Historical proposed design; not every component ran in V25. “Release/ship” did not mean clinical deployment. Current physician gold is scored only after output freeze.

The decision and the available evidence

Assignment evidence	Implementation choice
An incoming patient message	Choose the next care setting
Three supplied disposition labels	Keep the assignment interface
50 CSV rows; the brief describes 20	Run and retain every supplied message
Clinical disagreement with supplied labels	Use physician review for clinical evaluation

CSV labels remain an archived discussion baseline. They do not select the model or define clinical correctness.

The frozen three-bucket contract

Bucket	Meaning
SELF_CARE	Guidance only; no clinician action now
ASYNC_PHYSICIAN	Asynchronous physician review
URGENT_ESCALATION	Same-day in-person care or emergency action

Medication refill → ASYNC_PHYSICIAN unless clear same-day or emergency red flags require escalation.

Return one exact bucket and a short rationale as JSON.

URGENT_ESCALATION merges two levels of urgency. Route agreement does not establish safe timing or transport.

Live demonstration

C01: routine symptoms

C02: symptoms requiring urgent escalation

C06: stable medication refill

Edit C06 to add chest symptoms; inspect the new trace

Each submission starts one independent provider call.

Open /stripped. Use a clearly labeled saved record if the live attempt cannot complete. Exact inputs: appendix 23.

One call through a visible workflow

01

Patient message

02

One Mastra step

03

**Native Anthropic
call**

04

**Disposition and
trace**

TypeScript reproduces all 50 frozen Fable request bodies byte for byte.

The trace exposes final text, request IDs, token counts, and latency.

Input edits clear results and prevent stale answers from appearing.

No automatic retry or fallback. Exported traces omit hidden reasoning and credentials.

A reference revision, with the original retained

Reference view	Fable agreement	What changed
Original physician v2	44/49	C25 excluded; original record preserved
Revised labels, same 49 cases	47/49	C32, C34, C38 accepted as self-care
Physician v3, all 50 cases	48/50	C25 accepted as urgent escalation

C22 and C47 remain ASYNC_PHYSICIAN.

C25 receives a three-bucket label; the review does not specify emergency versus same-day care.

Single physician reassessment after viewing outputs; unblinded development evidence. Exact case text: appendices 24–25.

Frozen model comparison

Same three-bucket protocol	Original v2 /49	Revised v3 /50	Median latency
Opus 5 low	42	44	2.30 s
Fable 5.1 low	44	48	3.80 s
Fable 5.1 max	42	46	5.35 s
Astra extra-high	43	47	5.25 s
Astra max	43	47	9.58 s

Astra and Fable differ only on C07, C19, and C47.

Fable agrees on C07/C19; Astra agrees on C47. Both miss C22.

One 50-case pass per setting. Astra max is the highest verified API setting; ultra was unsupported. Exact C07/C19 messages: appendix 24; C47: slide 3.

Reference overtriage: three corrected self-care labels

Case	Physician correction	Separate advice requirement
C32: recent ear pain	ASYNC → SELF_CARE	Watchful waiting with reassessment criteria
C34: seasonal allergies	ASYNC → SELF_CARE	Check assumptions in medication advice
C38: improving sprain	ASYNC → SELF_CARE	Include pregnancy precautions for ibuprofen

The prior reference has false positives for clinician action; Fable is a true negative after correction. C38's frozen rationale omits pregnancy precautions. Correct routing does not resolve that omission.

Physician corrections apply to these cases. Rationale quality is a separate, unscored endpoint. Exact patient messages: appendix 25.

Clinical quality assurance uses separate endpoints

Question	How to evaluate it
Was the care setting appropriate?	Versioned physician reference; deterministic scoring
Was necessary clinician action missed?	False negatives with a defined positive endpoint
Was escalation unnecessary?	False positives; distinguish urgency from any review
Was the rationale appropriate?	Independent review of assumptions, medication advice, and precautions

The prototype has no judge call. These are evaluation dimensions, not added runtime steps.

Failure modes and evidence limits

Failure mode	Current evidence or required work
A stale answer appears under a new message	Verified edit cancellation and message matching
Reference correction appears to improve the model	Retained v2 scores and explicit v3 reconciliation
A correct route includes unsafe advice	C38 illustrates the need for separate rationale review
Urgent routing hides timing or transport	Requires an explicit operational policy
Known cases appear to be independent validation	New blinded cases and reviewers still required

Software checks and retrospective agreement do not establish clinical readiness.

First 30 days: one reviewed improvement

When	Artifact and accountable owner	Decision before the next step
Week 1	Routing and timing policy Clinical lead + care operations	Can independent clinicians apply the same boundaries?
Week 2	New cases and blinded physician labels Clinical evaluation lead	Are error definitions and acceptance limits fixed before inference?
Week 3	Paired test of missing-context handling Applied AI + clinical review	Does it reduce missed action within escalation and workload limits?
Week 4	One reviewed V0 and a reproducible report Engineering + clinical safety	Continue, revise or stop against the agreed criteria.

Proposed sequence. Week 4 produces a reviewed change and an evidence-based decision on its use.

Months 2–6: evidence for supervised use

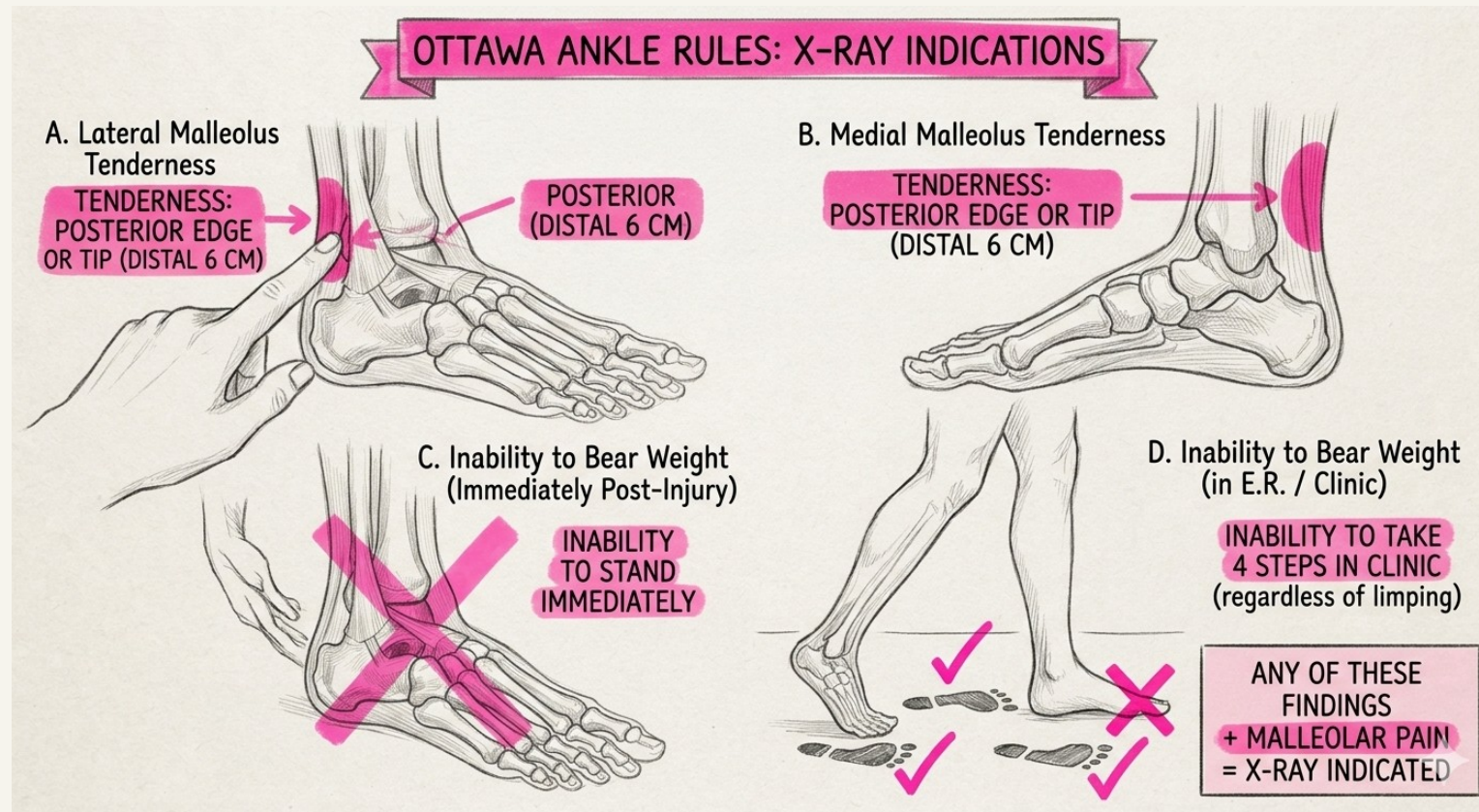
Horizon	Engineering outcome	Evidence before expansion
Months 2–3	A shared evaluation suite Owner: clinical evaluation + engineering	Any engineer can reproduce the comparison. Severe misses and advice quality remain separate.
Months 3–4	Observation alongside current care Owner: care operations + safety	Physicians retain decisions. Measure overrides, delays and clinician workload.
Months 4–6	One bounded supervised workflow Owner: clinical lead + engineering	Expand only within agreed safety limits, with incident review and a tested rollback.

First decision: when does missing information justify a question, and when does it require clinician review?

Proposed horizons depend on evidence and approved data access. Timing remains a separate validation task.

Appendix: C22 and Ottawa ankle assessment

Discussion illustration. Not part of the scored C22 input.



Weight-bearing criterion: unable to take four steps both immediately after injury and at assessment.
Illustration simplifies the rule. Use the cited clinical criteria.

Appendix: taxonomy changes the score

35/49

Five-way Opus, original exact-route score

46/49

Same outputs, collapsed to three buckets

Eleven disagreements disappear through mapping alone.

With revised v3 labels, this historical run scores 48/50; it used a different prompt.

Mapping and reference revision are separate changes. Neither changes the frozen predictions.

Appendix: assignment coverage

Take-home objective	Delivered evidence
Working disposition prototype	/stripped: three exact buckets and one call
Data and evaluation	All 50 messages; frozen runs; versioned physician scoring
Technical explanation	TypeScript/Mastra workflow and inspectable traces
Discuss tradeoffs and next steps	Case review, evidence limits, independent validation plan
Presentation and demonstration	15 main slides; 7-minute demo; 13 appendix slides

The project exceeded the original timebox. A timed rehearsal and final submission remain presenter responsibilities.

Appendix: frozen provider prompt

Choose exactly one bucket for the patient message:

1. SELF_CARE
2. ASYNC_PHYSICIAN
3. URGENT_ESCALATION

Medication refill → ASYNC_PHYSICIAN unless clear emergency/same-day red flags → URGENT_ESCALATION.

Return only JSON with "disposition" (one exact bucket name) and "rationale" (short).

Appendix: reproduce and inspect the evidence

Launch /stripped using the repository README.

Run: `node scripts/score-physician-adjudication-v3.mjs`

Inspect the v3 workbook for all 50 cases and Astra/Fable disagreements.

The original evaluation workbook and v2 scorecards remain available.

Frozen model outputs → versioned physician reference → offline scorecards.

Experimental appendix: timing within the assigned setting

Fixed care setting	Numbered timing choices	Agreement
Async physician review	1. Prompt physician review 2. Routine physician review	11/16
Urgent in-person care	1. Emergency action now 2. In-person assessment today	20/24

Inherited 48/50 three-bucket agreement includes C25 as correct urgent routing.

Finer agreement: 31/40 after correct setting decisions and 38/49 overall.

41 new calls and nine self-care cases. Only C25's emergency-versus-same-day label is unscored. Exact C25 message: appendix 24.

Experimental appendix: timing errors and policy conflicts

Cases	Model timing versus reference	Interpretation
C12, C16, C28, C44	Same-day instead of emergency	Timing undertriage
C13	Routine instead of prompt async	Priority undertriage
C06, C18, C24, C46	Prompt instead of routine async	Refill policy conflict

The prompt prioritizes refills. Four reference labels require routine review.

C22 and C47 remain first-stage self-care errors. Timing cannot correct them.

All nine subtype disagreements count as misses. Timing policy needs clinical review before operational use. Exact subtype-case messages: appendices 23, 26–28.

Case text: demonstration inputs

Case	Exact patient message
C01	Hi, I've had a runny nose and a mild sore throat for about 2 days. No fever, I'm still eating and drinking fine. Should I be worried?
C02	45M. I've had crushing pressure in the center of my chest for the last 30 minutes, it's going into my left arm and I feel sweaty and a bit sick to my stomach. I have high cholesterol.
C06	Could I get a refill on my losartan? I've been on 50mg once a day for about 2 years, my blood pressure is well controlled, last home reading was 122/78.

Verbatim synthetic CSV messages. Labels remain separate; these pages do not change the frozen inputs or scorecards.

Case text: model disagreements and urgent review

Case	Exact patient message
C07	My 3-year-old has had a fever up to 102F for 2 days. He's still drinking okay and pretty playful in between the fevers. No rash that I can see.
C19	I've been dizzy on and off for a few days. The room spins when I roll over or turn my head in bed, but it settles. No chest pain, no weakness, no headache.
C25	42F. I flew back from Europe two days ago and now my left calf is painful, swollen, and warm compared to the other one.

Verbatim synthetic CSV messages. Labels remain separate; these pages do not change the frozen inputs or scorecards.

Case text: corrected self-care references

Case	Exact patient message
C32	My 8-year-old says her right ear has hurt since last night. No fever, she ate breakfast fine and went to school.
C34	27F. My seasonal allergies are back — sneezing, itchy eyes, stuffy nose. What over the counter option do you recommend?
C38	22F. Following up on my ankle sprain from last week — it's improving. Just wondering how much ibuprofen I can safely take and for how long.

Verbatim synthetic CSV messages. Labels remain separate; these pages do not change the frozen inputs or scorecards.

Case text: additional refill timing cases

Case	Exact patient message
C18	Can I refill my albuterol inhaler? I use it maybe once a week, no bad attacks recently.
C24	33F. Could I get a refill on my birth control? I've been on the same one for three years with no problems.
C46	30M. Requesting a refill of my finasteride, 1mg daily. No side effects, been on it about a year.

Verbatim synthetic CSV messages. Labels remain separate; these pages do not change the frozen inputs or scorecards.

Case text: timing disagreements

Case	Exact patient message
C12	68F with COPD. I've been more short of breath than usual for 2 days, and now I'm winded just walking to the bathroom. My phlegm has turned green.
C13	55M. My right hand has been numb and tingly for a few days, worse at night. No weakness anywhere, no face drooping, my speech is normal.
C16	My 6-month-old has a fever of 100.9, she's fussy, she's had fewer wet diapers today than normal, and she doesn't seem interested in feeding.

Verbatim synthetic CSV messages. Labels remain separate; these pages do not change the frozen inputs or scorecards.

Case text: urgent timing disagreements

Case	Exact patient message
C28	29F, about 8 weeks pregnant. I started bleeding this morning, more than spotting, with cramping in my lower belly.
C44	68M. Both my ankles have been swelling for a couple weeks and it's gotten worse. The last two nights I've had to sit up to breathe because lying flat makes me short of breath.

Verbatim synthetic CSV messages. Labels remain separate; these pages do not change the frozen inputs or scorecards.